

A380X second ISIS: Standby Navigation Display (SND)

The lower ISIS screen, black until now, as the SND of the A380 FCOM: heading rose, position, waypoint list, DIR TO and FIX

Branch	pr/31-a380x-isis-snd
Aircraft	A380X (ISIS 2)
Commits	9dff226cb feat(a380x/isis): second ISIS as Standby Navigation Display (SND)
Base	upstream master
Status	CI checks pass; merged into develop (7b67b1c02); checked in the local browser harness against the FCOM figures; not tested in the sim yet.

Purpose

The A380 has two ISIS: one Standby Flight Display (SFD) and one Standby Navigation Display (SND) (FCOM DSC-34-10-20). FBW had the second one commented out in panel.cfg.

Specification (A380 FCOM DSC-34-10-20-30 and -50)

Part	Behaviour
Navigation	heading rose 240° (every 5°, numbers every 30°, markers at 45° and 90°), heading from ADIRU 3 (1), TRU in true reference, track diamond, GS, position with its source; HDG and PPOS flags
MENU and SET/SEL	INSERT WPT, INSERT FIX, EDIT WPT / FIX, CLEAR WPT / FIX (cyan); turn = scroll or value, push = select or next field; MENU again = exit
Waypoint list	FROM / TO / NEXT truncated to the minutes, TO -> NEXT true bearing and distance; TO WPT bearing (° or T) and distance; desired track, deviation bar and scale (5 / 2.5 NM)
DIR TO	LS/DIR TO pb: to a waypoint of the list (activates the navigation) or to new coordinates; sequencing abeam
FIX	distance to go, magenta bearing pointer, coordinates at the bottom of the list
Buttons, power	MODE: SND off / on; + / -: brightness; DC ESS, or DC HOT 1 above 50 kt

Simplifications (aircraft README, ATA 34)

- No SFD / SND reconfiguration, no power-up self-test, no FIX on the SFD.
- Position source always GPIR: FBW has no MMR / GPIRS model, the ADIRU position behaves as a hybrid one.
- Coordinate entry, list size (10) and sequencing: not described in the FCOM (its procedure figures stop at the first frame).

Tests

SndNavigation.spec.ts (10): geodesy, list and sequencing, menu flows (insert with the knob, scroll, exit, DIR TO both ways, clear), FIX.

PR description (from the PR draft)

From `pr-drafts/31-a380x-isis-snd.md`, section "Summary of Changes" onwards.

Summary of Changes

The A380 has two ISIS, one Standby Flight Display (SFD) and one Standby Navigation Display (SND) (FCOM DSC-34-10-20). FBW had the second one switched off. It is now a new FSComponent instrument, `A380X/SND`, on the lower ISIS screen, laid out on the FCOM figures (DSC-34-10-20-30):

- **Navigation:** heading rose (240°, graduated every 5°, numbers every 30°, fixed markers at 45° and 90°), yellow heading index, heading source IR3 (IR1 when ADIRU 3 has none), TRU in true reference (MAG/TRUE pb or extreme latitude, as the DMC), green track diamond, ground speed, aircraft position with its source; amber HDG (replaces the rose) and PPOS flags.
- **Waypoint list** (MENU pb, SET/SEL knob, FCOM DSC-34-10-20-50): INSERT WPT, INSERT FIX, EDIT WPT / FIX, CLEAR WPT / FIX in cyan; coordinates entered field by field with the knob; the list FROM / TO / NEXT (coordinates truncated to the minutes), the TO -> NEXT true bearing and distance, TO WPT bearing and distance, the desired track with the deviation bar and scale (large lines 5 NM, small 2.5 NM).
- **DIR TO** (LS/DIR TO pb): to a waypoint of the list (which activates the navigation) or to new coordinates; the TO waypoint is sequenced abeam.
- **FIX:** distance to go, magenta bearing pointer, coordinates at the bottom of the list.
- **MODE pb:** SND off / on; + and - pb: brightness. Power: DC ESS, or DC HOT 1 above 50 kt (as the SFD).
- Cockpit behaviour for the second ISIS (PUSH_ISIS_2_MODE / PLUS / MINUS / MENU / LS_DIR, KNOB_ISIS_2_SET_SEL: new template `FBW_ISIS_SND_Knob_Template`); `panel1.cfg` VCockpit11 gauge; `mach.config.js`.
- Simplifications listed in the aircraft README (ATA 34): no SFD/SND reconfiguration (MODE only switches the SND off and on), no power-up self-test, no FIX on the SFD, position source always GPIR (no MMR / GPIRS model), and the coordinate entry, the list size (10) and the sequencing are not described in the FCOM (its procedure figures stop at the first frame).
- 10 unit tests (`SndNavigation.spec.ts`): geodesy, list and sequencing, menu flows, FIX).

Cockpit API Changes

New H events: `A32NX_ISIS_2_{MODE|PLUS|MINUS|MENU|LS_DIR}_PRESSED`,
`A32NX_ISIS_2_KNOB_{CLOCKWISE|ANTI_CLOCKWISE|PRESSED}`.

Screenshots (if necessary)

TO ADD (from the sim): the SND in flight; the menu; the list with DIR TO active; the FIX.

References

- A380 FCOM DSC-34-10-20-10 (standby instruments), -30 (ISIS/SND: overview, indications, waypoints, FIX, position), -50 (how to use the MENU and SET/SEL in SND mode, insert a waypoint, DIR TO).

Discord username (if different from GitHub): **TO ADD**

Testing instructions

1. Power the aircraft: the lower ISIS shows the SND (heading rose, GS, IR3, GPIR3 position).
2. MODE: off, again: on; + / -: brightness.
3. MENU > INSERT WPT > push; set the coordinates with SET/SEL (turn = value, push = next field), push to the end. Insert a second one.
4. LS/DIR TO > push: FROM / TO / NEXT, TO WPT bearing and distance, the course arrow and the deviation bar.

5. MENU > INSERT FIX: FIX distance, magenta pointer, FIX line at the bottom of the list. EDIT / CLEAR WPT / FIX.
6. `npx vitest run fbw-a380x/src/systems/instruments/src/SND` (10 tests).